

Semantic Structure Predicts Exact Boolean Formula Complexity

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Abstract

We ask how much exact Boolean formula complexity is predictable from function semantics rather than formula syntax. Every one of the 65,536 four-input Boolean functions is synthesized exactly in NAND, NOR, and AND/OR/NOT formula languages. A tie-aware nearest-neighbor model uses only truth-table invariants: signed bias, sorted influences, Walsh–Fourier energy by degree, algebraic degree, and input-permutation stabilizer size. After removing every training function with the test descriptor, the model explains 77.7%, 77.7%, and 84.3% of exact minimum-size variance in the three languages. The best descriptor-measurable predictions explain 89.3%, 89.3%, and 93.1%, respectively. No matched permutation reaches an observed result ($p = 1/1001 < 0.001$). An order-invariant reanalysis of all 256 three-input functions yields lower strict values of 53.8%, 53.8%, and 58.9%; the four-input result is stable across neighbor thresholds and coordinate scalings. Elementary semantics therefore contains substantial, scalable information about exact representation cost, while imperfect cross-language rank stability confirms that cost is not intrinsic to the function alone.

1 Problem and formal setup

Takeaway. Meaning strongly correlates with and predicts representation cost, but does not completely dictate it: even the full descriptor leaves a nonzero within-class residual, and the chosen gate language still matters.

Let $\mathcal{B}_n = \{f : \{0, 1\}^n \rightarrow \{0, 1\}\}$, with $n \in \{3, 4\}$. A formula language $\mathcal{L}$ consists of variables $x_0, \dots, x_{n-1}$ and a fixed gate basis. Formulas are rooted trees, and size $|e|$ is the number of gates. This basis-relative notion descends from classical switching synthesis [1] and remains central in modern exact synthesis [6, 7].

Definition 1 (Exact formula complexity). For evaluation map $\text{eval}_{\mathcal{L}} : E_{\mathcal{L}} \rightarrow \mathcal{B}_n$, define

$$K_{\mathcal{L}}(f) = \min\{|e| : \text{eval}_{\mathcal{L}}(e) = f\}. \quad (1)$$

The three languages are NAND, NOR, and ordered AND/OR/NOT. They are functionally complete but assign different costs to the same truth table. For $f \in \mathcal{B}_n$, let

$$\psi_n(f) = (b(f), \text{sort}(I_1(f), \dots, I_n(f)), E_0(f), \dots, E_n(f), \deg_{\mathbb{F}_2}(f), |\text{Stab}(f)|) \in \mathbb{R}^{2n+4}. \quad (2)$$

Here b is signed output bias, I_i is variable influence, E_d is Walsh–Fourier energy at degree d , and the stabilizer is under input permutations. These are computed from the truth table and contain no gate or formula data. Influence and Fourier summaries have classical connections to circuit structure and learnability [2, 3]; O’Donnell and Servedio directly relate influence to decision-tree size [4].

Proposition 2 (Input-renaming invariance). For every $\pi \in S_n$, language $\mathcal{L}$, and function f ,

$$K_{\mathcal{L}}(f \circ \pi) = K_{\mathcal{L}}(f), \quad \psi_n(f \circ \pi) = \psi_n(f). \quad (3)$$

Proof. Relabeling every leaf is a size-preserving bijection of formulas. Bias, sorted influences, Fourier energy by degree, algebraic degree, and stabilizer cardinality are coordinate-permutation invariants. $\square$

2 Exact synthesis

Let $M_{\ell} \subseteq \mathcal{B}_n$ be the functions whose minimum formula size is exactly ℓ , with M_0 the variable truth tables. If $\mathcal{O}_r$ is the set of r -ary gates, define the candidate layer

$$R_{\ell} = \bigcup_{r \geq 1} \bigcup_{o \in \mathcal{O}_r} \bigcup_{\ell_1 + \dots + \ell_r = \ell - 1} \{o(f_1, \dots, f_r) : f_j \in M_{\ell_j}\}. \quad (4)$$

Theorem 3 (Minimum-layer recurrence). *For every $\ell \geq 1$,*

$$M_\ell = R_\ell \setminus \bigcup_{j < \ell} M_j. \quad (5)$$

Consequently, the first layer containing f gives $K_{\mathcal{L}}(f)$ exactly.

Proof. Every size- ℓ formula has a root gate and subformula sizes summing to $\ell - 1$. In a minimum formula, each subformula must itself be minimum for the function it computes; otherwise replacing it would shorten the whole formula. Thus every minimum formula occurs in R_ℓ , and removing earlier layers leaves precisely the new minima. The reverse inclusion follows by construction. $\square$

Vectorized evaluation of (5) reaches every four-input truth table. As independent checks, the implementation reproduces all previously counted three-input minima and verifies the all-function duality $K_{\text{NAND}}(f) = K_{\text{NOR}}(\neg f(\neg x))$. Table 1 summarizes the complete universes.

Universe	Functions	S_n orbits	Descriptor classes	Maximum minima
$n = 3$	256	80	36	NAND/NOR 14
$n = 4$	65,536	3,984	675	NAND/NOR 26; A/O/N 19

Table 1: Complete synthesis scale. Four-input mean minima are 12.231 gates for NAND/NOR and 10.141 for AND/OR/NOT.

3 Prediction and falsification protocol

Descriptor coordinates are standardized. For a test function f , let $T(f)$ be the permitted training functions and let $r_k(f)$ be the distance to its k th nearest member. To remove arbitrary truth-table ordering when distances tie, define

$$N_k^*(f) = \{g \in T(f) : \|\psi_n(g) - \psi_n(f)\|_2 \leq r_k(f)\}, \quad \hat{K}_{\mathcal{L}}(f) = \frac{1}{|N_k^*(f)|} \sum_{g \in N_k^*(f)} K_{\mathcal{L}}(g), \quad (6)$$

with $k = 7$. Thus at least seven neighbors are used, including the complete boundary tie shell.

Three nested protocols remove (i) only f ; (ii) the complete S_n orbit of f ; or (iii) the complete descriptor fiber $\psi_n^{-1}(\psi_n(f))$. The strict protocol therefore contains neither an input-renamed copy nor any function with the same descriptor. Performance is

$$R^2 = 1 - \frac{\sum_f (K_{\mathcal{L}}(f) - \hat{K}_{\mathcal{L}}(f))^2}{\sum_f (K_{\mathcal{L}}(f) - \bar{K}_{\mathcal{L}})^2}. \quad (7)$$

The descriptor ceiling predicts each function by its descriptor-class mean. It is the largest possible R^2 for a predictor measurable only with respect to these exact classes, since the class mean minimizes within-class squared error. The matched null preserves signed bias, algebraic degree, and stabilizer size. Within each exact stratum z , complexity labels are permuted and the fixed strict plan is reevaluated, testing

$$H_0 : K_{\mathcal{L}} \perp\!\!\!\perp \psi_n \mid z, \quad p = \frac{1 + \#\{b : R_b^2 \geq R_{\text{obs}}^2\}}{B + 1}, \quad B = 1000. \quad (8)$$

The plus-one correction prevents zero Monte Carlo p -values [8].

4 Results

The main result is the descriptor-held-out column of Table 2. Semantics explains 77.7% of exact NAND/NOR variance and 84.3% of AND/OR/NOT variance without an exact semantic twin in training.

Language	Function R^2	Orbit R^2	Strict R^2	Ceiling R^2	RMSE	Null 95%
NAND	.890	.851	.777	.893	1.505	[.038, .048]
NOR	.890	.851	.777	.893	1.505	[.038, .048]
AND/OR/NOT	.929	.906	.843	.931	1.031	[.047, .056]

Table 2: Four-input prediction. No matched permutation reaches an observed strict result, so $p = 1/1001 < 0.001$ in every language.

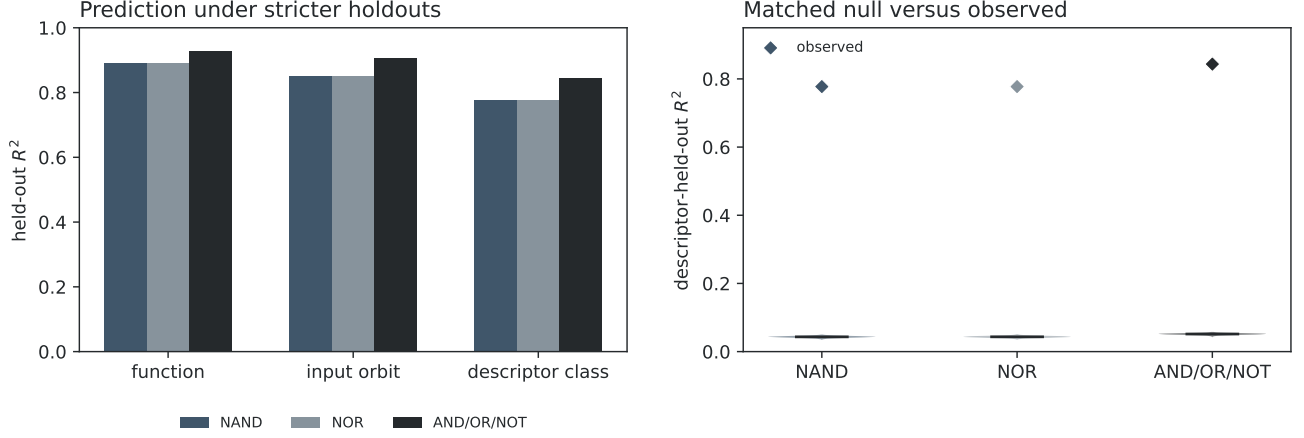


Figure 1: Leakage and null controls. Prediction attenuates under stricter holdouts (left), yet strict observations remain far above bias/degree/symmetry- matched permutation distributions (right).

The same tie-aware protocol on three inputs gives strict R^2 values .538, .538, and .589, compared with .777, .777, and .843 at four inputs. Thus the stronger signal is not an artifact of changing the tie rule. Table 3 shows that bias, degree, and symmetry alone carry essentially no strict predictive power, whereas influence and Fourier coordinates are independently informative.

Features	NAND R^2	NOR R^2	AND/OR/NOT R^2
Bias, degree, symmetry	.002	.002	-.059
Influences only	.669	.669	.819
Fourier energy only	.666	.666	.828
Without influences	.637	.637	.636
Without Fourier	.791	.791	.806
Full descriptor	.777	.777	.843

Table 3: Four-input descriptor ablations under strict holdout. The slight NAND/NOR gain after removing Fourier coordinates indicates Euclidean-distance dilution, not absence of Fourier signal.

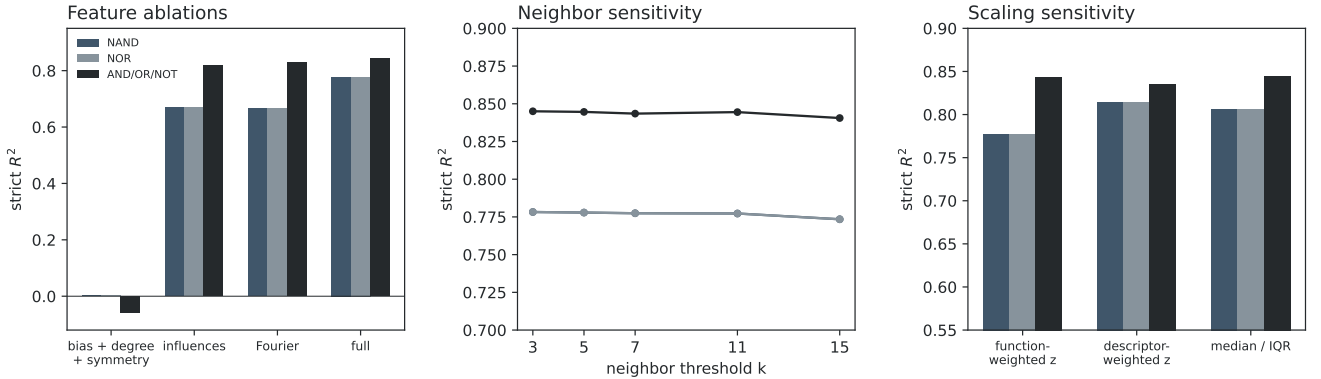


Figure 2: Feature and protocol robustness. Across $k \in \{3, 5, 7, 11, 15\}$, strict R^2 remains .773–.778 for NAND/NOR and .841–.845 for AND/OR/NOT. Alternative scaling keeps strict R^2 at .806–.815 for NAND/NOR and .836–.844 for AND/OR/NOT.

The representation languages remain distinct. Spearman correlations of minimum cost are .664 for NAND versus NOR and .860 for each comparison with AND/OR/NOT. Identical aggregate NAND/NOR prediction scores arise

from their duality and the invariant, tie-aware evaluation; their functionwise rankings are not identical.

5 Interpretation and limitations

The complete four-input experiment strengthens a narrow claim: elementary, syntax-free properties of a Boolean function contain substantial information about exact minimal representation cost beyond input renaming, descriptor duplication, and the matched effects of bias, degree, and symmetry. The strict predictor also approaches the descriptor ceiling: it recovers .777 of a maximum .893 for NAND/NOR and .843 of .931 for AND/OR/NOT. The result is complementary to analytic lower-bound programs based on Fourier concentration or related function measures [2, 5]; it is finite, predictive, and exact rather than an asymptotic lower bound.

It does not show that “meaning determines difficulty.” Identical descriptors retain within-class cost variation, cross-language rankings are imperfect, and changing coordinate weighting changes the exact prediction. The universe is complete but still limited to four inputs, three related classical bases, and a hand-designed descriptor. Stronger follow-up tests should use selected five-input families or exact databases, substantially different computational models, and learned metrics evaluated under the same fiber holdout.

Reproducibility. The checked-in experiment generates every exact target, fold, ablation, robustness check, and all 1,000 null draws (seed 0). It uses no model APIs or cloud services.

References

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